# SIMATS ENGINEERING
## ASSESSMENT TOOL 1: Analytical Problem Solving

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|---|---|---|---|
| **COURSE NAME:** | Cloud Computing and Big Data Analytics | **COURSE CODE:** | CSA15 |

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**COURSE OUTCOME COVERED**
**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

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**Code of Conduct**
I _Lavanya.M_(192321051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

**Assessment Tasks**
**Analytical Problem Solving**

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

**Analytical Problem Solving Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 – Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Problem Interpretation | 20% | Interprets all technical requirements accurately | Interprets most requirements correctly | Partial interpretation | Misinterprets the problem |
| Analytical Reasoning | 25% | Strong reasoning with clear cause-effect analysis | Good reasoning with minor gaps | Basic analysis only | Weak or no analysis |
| Method Selection | 20% | Chooses highly appropriate virtualization and security concepts | Chooses mostly suitable concepts | Partially suitable concepts | Inappropriate approach |
| Solution Quality | 20% | Provides feasible and technically sound solution | Reasonable solution with minor issues | Basic solution with limitations | Unclear or invalid solution |
| Presentation of Analysis | 15% | Well-structured and technically precise | Generally clear | Somewhat unclear | Difficult to follow |

**Total Marks Awarded:**

**1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications (10 Marks)**

**Introduction**

A hypervisor is virtualization software that enables multiple Virtual Machines (VMs) to run on a single physical server. Hypervisors are classified into **Type-1 (Bare-Metal)** and **Type-2 (Hosted)** based on their architecture.

| Feature | Type-1 Hypervisor | Type-2 Hypervisor |
|---|---|---|
| Installation | Runs directly on hardware | Runs on top of a host operating system |
| Performance | Very high | Lower due to host OS overhead |
| Security | More secure | Less secure because host OS can be attacked |
| Resource Usage | Efficient | Consumes extra resources |
| Examples | VMware ESXi, Microsoft Hyper-V, Xen | Oracle VirtualBox, VMware Workstation |

**Performance Analysis**

- Type-1 hypervisors communicate directly with hardware, resulting in lower latency and higher throughput.
- Type-2 hypervisors depend on the host operating system, increasing CPU and memory overhead.
- Mission-critical applications require maximum uptime and fast response times, making Type-1 more suitable.

**Security Analysis**

- Type-1 has a smaller attack surface because there is no general-purpose operating system between hardware and VMs.
- Type-2 inherits vulnerabilities from the host operating system, increasing security risks.
- Type-1 provides stronger VM isolation and secure resource allocation.

**Manageability Analysis**

- Type-1 supports centralized management, live migration, snapshots, clustering, and automatic failover.
- Type-2 is mainly intended for software testing and desktop virtualization rather than enterprise cloud environments.

**Justification**

For a cloud datacenter hosting banking, healthcare, or government applications, **Type-1 hypervisors are the better choice** because they provide:

- Better performance
- Stronger security
- Higher availability
- Easier centralized management
- Enterprise scalability

**Conclusion**

Type-1 hypervisors are ideal for mission-critical cloud infrastructures because they deliver high performance, strong isolation, and enterprise-grade reliability, whereas Type-2 hypervisors are more appropriate for development and testing.

## 2. Workload Isolation Using Virtualization Architecture and Attack Surfaces

### Introduction

Cloud providers often host multiple organizations on the same physical servers. Virtualization enables multiple isolated VMs to securely share hardware resources.

### Virtualization-Based Isolation

#### 1. Hypervisor Isolation

Each VM operates independently with its own operating system.

#### 2. Resource Isolation

CPU, RAM, storage, and network resources are allocated separately to each VM.

#### 3. Virtual Network Isolation

Different departments such as Finance, Healthcare, and Student Portals are placed on separate VLANs or virtual networks.

#### 4. Storage Isolation

Each workload uses separate virtual disks and storage permissions to prevent unauthorized access.

### Example

- Finance VM → Payroll and financial records
- Healthcare VM → Patient medical records
- Student Portal VM → Student information

Even though all run on the same server, virtualization ensures that one VM cannot directly access another VM's data.

### Major Attack Surfaces

#### 1. Hypervisor Attack

- Attackers exploit vulnerabilities in the hypervisor.
- Can compromise multiple VMs simultaneously.
- Prevented by regular patching and secure hypervisor configuration.

#### 2. VM Escape Attack

- Malicious software inside one VM escapes into the hypervisor.
- May gain access to neighboring VMs.
- Prevented through strict isolation and updated virtualization software.

### Security Measures

- Role-Based Access Control (RBAC)

- Multi-Factor Authentication
- Encryption of virtual disks
- Network segmentation
- Continuous monitoring

**Conclusion**

Virtualization architecture enables secure workload isolation while efficiently utilizing hardware resources. Protecting the hypervisor and preventing VM escape attacks are essential for maintaining cloud security.

**3. VM Host Utilization Analysis and Corrective Actions (10 Marks)**

**Given Snapshot**

| Resource | Value |
|---|---|
| CPU Utilization | 92% |
| Memory Utilization | 88% |
| Storage I/O Latency | 35 ms |
| Average VM Boot Time | 170 s |

**Analysis**

**CPU Utilization (92%)**

- Extremely high CPU usage
- Indicates CPU contention among multiple VMs
- Applications may experience slow response times.

**Memory Utilization (88%)**

- Very high RAM consumption
- Hypervisor may begin swapping memory to disk
- Leads to slower VM performance.

**Storage I/O Latency (35 ms)**

- Storage delay is much higher than the recommended (<10 ms)
- Indicates storage bottlenecks
- Causes slow file access and database operations.

**VM Boot Time (170 s)**

- Normal VM boot time is usually below one minute.
- Long boot times indicate overloaded CPU, memory, or storage.

**Likely Bottlenecks**

- CPU overcommitment
- Insufficient RAM
- Slow storage subsystem

**Corrective Actions**

**CPU Optimization**

- Migrate some VMs to another host.
- Increase CPU cores.
- Enable load balancing.

**Memory Optimization**

- Upgrade RAM.
- Enable Dynamic Memory allocation.
- Remove idle VMs.

**Storage Improvement**

- Replace HDD with SSD or NVMe storage.
- Use faster SAN storage.
- Optimize storage caching.

**VM Optimization**

- Remove unused snapshots.
- Allocate appropriate VM resources.
- Use thin provisioning carefully.

**Conclusion**

The system suffers from CPU, memory, and storage bottlenecks simultaneously. Resource balancing, hardware upgrades, and VM optimization will significantly improve performance and reduce boot times.

## 4. Software Defined Datacenter (SDDC) vs Traditional Datacenter (10 Marks)

### Introduction

A **Software Defined Datacenter (SDDC)** virtualizes compute, storage, and networking resources, allowing them to be managed entirely through software.

### Traditional Datacenter

- Hardware configured manually
- Fixed networking
- Dedicated storage devices
- Slow provisioning
- Limited scalability

### Software Defined Datacenter

### Compute Virtualization

- Physical servers host multiple VMs.
- Resources are dynamically allocated.
- Live migration improves availability.

### Storage Virtualization

- Multiple storage devices appear as one logical storage pool.
- Automatic storage allocation.
- Better utilization and disaster recovery.

### Network Virtualization

- Virtual switches and Software Defined Networking (SDN).
- Rapid network configuration.
- Network policies managed centrally.

**Agility Improvements**

| Traditional | Software Defined Datacenter |
|---|---|
| Manual provisioning | Automated provisioning |
| Weeks to deploy servers | Minutes to deploy VMs |
| Hardware-dependent | Software-controlled |
| Difficult scaling | Automatic scaling |
| Manual management | Centralized management |

**Benefits**

- Faster deployment
- Better resource utilization
- Lower operational cost
- High availability
- Simplified disaster recovery
- Easier automation

**Conclusion**

SDDC greatly improves agility by virtualizing compute, storage, and networking. Organizations can rapidly provision resources, automate operations, and scale services efficiently compared to traditional datacenters.

## 5. Multi-Cloud Identity Federation and Secure Privacy Design (10 Marks)

### Introduction

Multi-cloud environments use services from multiple cloud providers. Identity mismatches occur when users have different credentials across providers, making authentication and authorization difficult.

### Problem Analysis

#### Identity Mismatch

- Different username formats
- Different authentication protocols
- Separate user databases
- Inconsistent access policies

#### Consequences

- Login failures
- Duplicate accounts
- Poor user experience
- Increased security risks
- Administrative complexity

### Federation-Based Solution

#### Central Identity Provider (IdP)

Use a single trusted Identity Provider to authenticate users.

#### Single Sign-On (SSO)

Users log in once and access multiple cloud providers without repeated authentication.

#### Federation Standards

- **SAML (Security Assertion Markup Language):** Secure exchange of authentication information.
- **OAuth 2.0:** Secure delegated authorization for applications.
- **OpenID Connect (OIDC):** Authentication layer built on OAuth 2.0.

### Privacy Design

- Encrypt identity data during transmission and storage.
- Use Multi-Factor Authentication (MFA).
- Implement Role-Based Access Control (RBAC).
- Share only the minimum required user attributes (principle of least privilege).

**Secure Identity Architecture**

User

↓

Identity Provider (IdP)

↓

SSO Authentication

↓

Cloud Provider A

Cloud Provider B

Cloud Provider C

**Benefits**

- One secure login
- Consistent identity across providers
- Improved security
- Reduced administrative effort
- Better privacy protection
- Simplified access management

**Conclusion**

Identity federation using **SAML**, **OAuth 2.0**, and **OpenID Connect**, combined with SSO, MFA, RBAC, encryption, and audit logging, provides a secure and scalable solution for resolving identity mismatches in multi-cloud environments while protecting user privacy.